

focus mechanism adjusts to the new distance of the subject. Conversely, manual focusing while panning requires a fair bit of skill, which might take you some time to develop.

### 1.2.6. Shutter Speed

The shutter is the mechanical assembly that controls the speed at which light is admitted to the CCD. This is a circular device that rotates, opening and shutting with each rotation. The speed of the shutter opening is what is known as shutter speed and is measured in rotations per second. The range of shutter speed operation will vary according to the model and make of camcorder. This can be from anything as slow as 1/60th of a second to as high as 1/15000th of a second! Recording of high speed motion requires higher (faster) shutter speeds. Recording in low light requires lower (slower) shutter speeds, since the shutter needs to remain open for much longer to admit a sufficient amount of light. If a high-speed motion scene like a car chase is recorded at a low shutter speed, the resulting video will have a blurred or trailing effect. For crisp and sharp video, you need to record it at higher shutter speeds. However, the question is, what if you need to record high-speed motion in low-light conditions? This is a trade off which can to some extent be alleviated by using the correct aperture rating or F-Stop.



### 1.2.7. Aperture

While the shutter speed controls the speed at which light is admitted, the aperture controls the amount of light that will reach the CCD. This can be imagined as a black window that opens and shuts continuously very fast (shutter) and a curtain (aperture) behind the window that controls the strength of the sunlight entering the room (CCD). The wider the iris (aperture), the more the amount of light that reaches the CCD. This in turn results in a smaller F-stop, which is the measure of exposure. Generally, the aperture size is fixed in most consumer camcorders, so look for a lower F-stop to give you better results in low light situations. The